The documentation and process conversion measures necessary to comply with this revision shall be completed by 30 November 1993

INCH-POUND

MIL-S-19500/108E
30 August 1993
SUPERSEDING
MIL-S-19500/108D
27 April 1972

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, THYRISTORS (CONTROLLED RECTIFIERS), SILICON, TYPES 2N682, 2N683, 2N685 THROUGH 2N692, 2N5206, JAN, AND JANTX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for PNPN silicon power, reverse-blocking triode, thyristors. Two levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

	I_O 1/ T_C = +65°C (180° condition angle)	I_{TSM} 2/ t = 7 ms	V_{GM}	T_{op}	T_{STG}	di/dt (repetitive)
	A	A	V	°C	°C	A/µs
Min				-65	-65	5
Max	16	150	5	+125	+150	

1/ This average forward current is for a maximum case temperature of +65°C, and 180 electrical degrees of conduction.
2/ Surge rating is non-recurrent and applies only with device in the conducting state. The peak rate of surge current must not exceed 100 amperes during the first 10 µs after switching from the "off" (blocking) state to the "on" (conducting) state. This time is measured from the point where the thyristor voltage has decayed to 90 percent of its initial blocking value.

1.4 Primary electrical characteristics (common to all types).

Limits	V_{TM}	I_H	V_{GT} 1/	I_{GT} 1/
	v(pk)	mA dc	V dc	mA dc
Min			0.25	
Max	2.0	50	3.0	80

1/ T_C = -65° to +125°C.

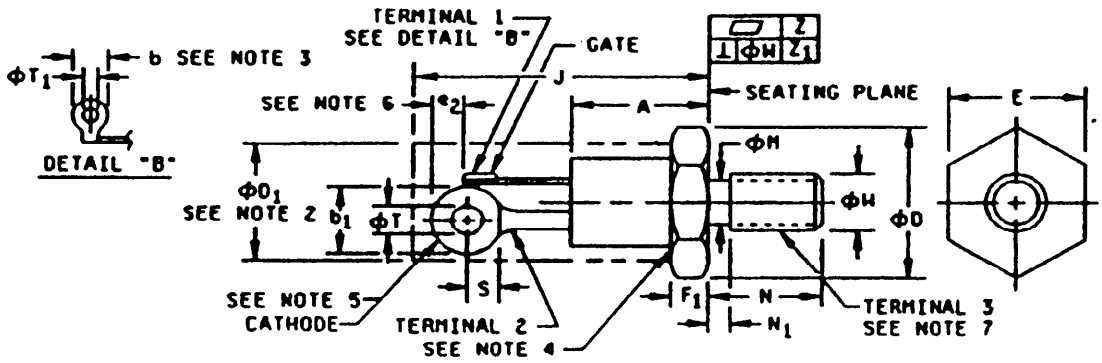
Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ES, 1507 Wilmington Pike, Dayton, OH 45444-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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Symbol	Dimension				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A		.550		13.97	
b	.115	.139	2.93	3.55	3
b1	.210	.300	5.34	7.62	3
ϕD		.650		16.51	
ϕD1		.543		13.81	2
E	.544	.563	13.82	14.30	
e2	.125		3.18		6
F1	.075	.200	1.91	5.08	4
J		1.193		30.30	2
ϕM	.220	.249	5.59	6.32	
N	.422	.453	10.72	11.50	
N1		.090		2.28	
S	.120		3.05		3
ϕT	.125	.165	3.18	4.19	
ϕT1	.060	.075	1.53	1.90	
ϕW	.25-28 UNF 2A				
Z		.002		.050	
Z1		.006		.152	

NOTES:
1. Metric equivalents are given for general information only.
2. Device contour except on hex head and noted terminal dimensions is optional within zone defined by ϕD1 and J, ϕD1 not to exceed actual E.
3. Contour and angular orientation of terminals 1 and 2 with respect to hex portion and to each other are optional.
4. Chamfer or undercut on one or both ends of the hexagonal portion are optional.
5. Square or radius on end of terminal is optional.
6. Minimum difference in terminal lengths to establish datum line for numbering terminals.
7. Dimension ϕW is pitch diameter of coated threads.

FIGURE 1. Physical dimensions of thyristor types 2N682, 2N683, 2N685 through 2N692, and 2N206.

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1.4.1 Individual ratings.

Type	V_{RRM} 1/	V_{DRM}	V_{RSM} 2/
	v(pk)	v(pk)	v(pk)
2N682	50	50	75
2N683	100	100	150
2N685	200	200	300
2N686	250	250	350
2N687	300	300	400
2N688	400	400	500
2N689	500	500	600
2N690	600	600	720
2N691	700	700	840
2N692	800	800	960
2N5206	1,000	1,000	1,200

1/ Values apply for zero or negative gate voltage (V_{GM}).
2/ Peak reverse voltage, nonrepetitive, t = 5.0 ms max.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

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3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500, and as follows:

V_{AA} - Anode power supply voltage (dc).

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figure 1 herein.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750 and MIL-S-19500. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor, but shall be retained on the initial container.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANTX level only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement JANTX level
3a	Condition F, 10 cycles
4	5,000 g's
7	Condition C for gross leaks
10	Not required
11	I_{RRM1}, I_{DRM1}, V_{GT1}, V_{TM}, and I_{GT1}
12d	Condition A, T_C = +125°C ±2°C; V_{DRM}, V_{RRM} = rated (see 1.4.1).
13	Subgroup 2 of table I herein, ΔI_{RRM} = 100 percent of initial value or +0.4 mA(pk), whichever is greater. ΔI_{DRM} = 100 percent of initial value or +0.4 mA(pk), whichever is greater.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein.

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4.4.2 Group B inspection (JANTX and JANTXV). Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX and JANTXV) of MIL-S-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps and footnotes of table II herein. Subgroups 3 and 6 shall be performed on a sample from the sublot containing the highest voltage rated devices in the lot.

Subgroup	Method	Condition
B3	1037	ΔT_J between cycles ≥ +100°C; $t_{on} = t_{off}$ ≥ 1 minute for 2,000 cycles.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps and footnotes of table II herein.

Subgroup	Method	Condition
C2	2036	Test condition A, weight = 10 lbs, application time = 15 seconds.
C6	1037	V_{CB} ≥ 10 V dc; ΔT_J between cycles ≥ +100°C, $t_{on} = t_{off}$ ≥ 1 minute for 6,000 cycles. No heat sink or forced air cooling on the device shall be permitted.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Reverse blocking current 2N682 through 2N692 and 2N5206	4211	AC method, bias condition D; f = 60 Hz; V_{RRM} = rated (see 1.4.1)	I_{RRM1}		2	mA(pk)
Forward blocking current 2N682 through 2N692 and 2N5206	4206	AC method, bias condition D; f = 60 Hz; V_{DRM} = rated (see 1.4.1)	I_{DRM1}		2	mA(pk)
Gate trigger voltage and current	4221	V_2 = V_D = 6 V dc; R_L = 50Ω; R_e = 20Ω maximum	V_{GT1} I_{GT1}		3 35	V dc mA dc
Forward "on" voltage	4226	I_{TM} = 50 A(pk) (pulse); pulse width = 8.5 ms; maximum; duty cycle = 2 percent maximum	V_{TM}		2	v(pk)
Holding current	4201	Bias condition D; V_{AA} = 24 V dc maximum; I_{TM} = I_{F1} = 1 A dc; I_T = I_{F2} = 100 mA dc; trigger voltage source = 10 V; trigger PW = 100 µs (minimum); R_2 = 20Ω	I_H		50	mA dc
Reverse gate current	4219	V_G = 5 V dc	I_G		250	mA dc
Subgroup 3						
High temperature operation:		T_C = +125°C				
Reverse blocking current 2N682 through 2N692 and 2N5206	4211	AC method, bias condition D; f = 60 Hz; V_{RRM} = rated (see 1.4.1)	I_{RRM2}		5	mA(pk)
Forward blocking current 2N682 through 2N692 and 2N5206	4206	AC method, bias condition D; f = 60 Hz; V_{DRM} = rated (see 1.4.1)	I_{DRM2}		5	mA(pk)

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
Gate trigger voltage	4221	T_C = +125°C; R_e = 20Ω maximum	V_{GT2}	.25		V dc
2N682		$V_2 = V_{DM}$ = 50 V dc; R_L = 100Ω				
2N683		$V_2 = V_{DM}$ = 100 V dc; R_L = 100Ω				
2N685		$V_2 = V_{DM}$ = 200 V dc; R_L = 100Ω				
2N686		$V_2 = V_{DM}$ = 250 V dc; R_L = 400Ω				
2N687		$V_2 = V_{DM}$ = 300 V dc; R_L = 400Ω				
2N688		$V_2 = V_{DM}$ = 400 V dc; R_L = 400Ω				
2N689		$V_2 = V_{DM}$ = 500 V dc; R_L = 400Ω				
2N690		$V_2 = V_{DM}$ = 600 V dc; R_L = 1 KΩ				
2N691		$V_2 = V_{DM}$ = 700 V dc; R_L = 1 KΩ				
2N692		$V_2 = V_{DM}$ = 800 V dc; R_L = 1 KΩ				
2N5206		$V_2 = V_{DM}$ = 1,000 V dc; R_L = 1 KΩ				
Low temperature operation:		T_C = -65°C				
Reverse blocking current	4211	AC method, bias condition D; f = 60 Hz; V_{RRM} = rated (see 1.4.1)	I_{RRM3}		2	mA(pk)
2N682 through 2N692 and 2N5206						
Forward blocking current	4206	AC method, bias condition D; f = 60 Hz; V_{DRM} = rated (see 1.4.1)	I_{DRM3}		2	mA(pk)
2N682 through 2N692 and 2N5206						
Gate trigger voltage and current	4221	$V_2 = V_D$ = 6 V dc; R_L = 50Ω; R_e = 20Ω maximum	V_{GT3} I_{GT2}		3 80	V dc mA dc
Subgroups 4 and 5						
Not applicable						
Subgroup 6						
Surge current	4066	I_{TSM} = 150 A(pk) (.5 sine wave); 10 surges per 1 minute; I_o = 16 A at rated V_{RRM}; T_C = +65°C; f = 60 Hz; surge duration = 7 ms, minimum				
Electrical measurements		See table II, steps 1, 2, 3, 4, 5, and 6.				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 7						

Exponential rate of voltage rise	4231	Bias condition D; T_C = +125°C
		dv/dt = 25 v/µs;
		repetition rate = 60 pps;
		test duration = 15 s;
		C = 1.0 µf; R_L = 50Ω
			V_D			

2N682		V_{AA} = 50 V dc		47		V dc
2N683		V_{AA} = 100 V dc		95		V dc
2N685		V_{AA} = 200 V dc		190		V dc
2N686		V_{AA} = 250 V dc		240		V dc
2N687		V_{AA} = 300 V dc		285		V dc
2N688		V_{AA} = 400 V dc		380		V dc
2N689		V_{AA} = 500 V dc		475		V dc
2N690		V_{AA} = 600 V dc		570		V dc
2N691		V_{AA} = 700 V dc		665		V dc
2N692		V_{AA} = 800 V dc		760		V dc
2N5206		V_{AA} = 1,000 V dc		950		V dc

Circuit-commutated turn-off time	4224	T_C = +120°C; I_{TM} = 10 A;
		t_{on} = 100 ±50 µs;
		di_1/dt = 5 A/µs minimum;
		di_1/dt = 8 A/µs maximum;
		reverse voltage at t_1 = 15 V minimum;
		repetition rate = 60 pps max;
		dv/dt = 20 V/µs; gate bias conditions:
		gate source voltage = 0 V;
		gate source resistance = 100Ω
			t_{off}			

2N682		$V_{DM} = V_{DRM}$ = 50 V(pk); V_{RRM} = 50 V maximum			30	µs
2N683		$V_{DM} = V_{DRM}$ = 100 V(pk); V_{RRM} = 100 V maximum			30	µs
2N685		$V_{DM} = V_{DRM}$ = 200 V(pk); V_{RRM} = 200 V maximum			30	µs
2N686		$V_{DM} = V_{DRM}$ = 250 V(pk); V_{RRM} = 250 V maximum			30	µs
2N687		$V_{DM} = V_{DRM}$ = 300 V(pk); V_{RRM} = 300 V maximum			30	µs
2N688		$V_{DM} = V_{DRM}$ = 400 V(pk); V_{RRM} = 400 V maximum			30	µs
2N689		$V_{DM} = V_{DRM}$ = 500 V(pk); V_{RRM} = 500 V maximum			40	µs
2N690		$V_{DM} = V_{DRM}$ = 600 V(pk); V_{RRM} = 600 V maximum			40	µs
2N691		$V_{DM} = V_{DRM}$ = 700 V(pk); V_{RRM} = 700 V maximum			60	µs
2N692		$V_{DM} = V_{DRM}$ = 800 V(pk); V_{RRM} = 800 V maximum			60	µs
2N5206		$V_{DM} = V_{DRM}$ = 1,000 V(pk); V_{RRM} = 1,000 V maximum			60	µs
Gate controlled turn-on time	4223	V_{AA} = 50 V dc for 2N682, 100 V dc for 2N683 through 2N692 and 2N5206; I_{TM} = 10 A dc; V_{GG} = 10 V dc; R_e = 25Ω; t_{p1} = 15 ±5 µs; 4 A/µs ≤ di_1/dt ≤ 200 A/µs.	t_{on}		5	µs

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Groups A, B, and C electrical end-point inspection measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Reverse blocking cutoff current	4211	AC method, bias condition D; f = 60 Hz; V_{RRM} = rated (see 1.4.1)	I_{RRM1}		2	mA(pk)
2.	Forward blocking current	4206	AC method, bias condition D; f = 60 Hz; V_{DMR} = rated (see 1.4.1)	I_{DRM1}		2	mA(pk)
3.	Reverse blocking cutoff current	4211	AC method, bias condition D; T_C = +125°C; f = 60 Hz; V_{RRM} = rated (see 1.4.1)	I_{RRM2}		5	mA(pk)
4.	Forward blocking current	4206	AC method, bias condition D; T_C = +125°C; f = 60 Hz; V_{DMR} = rated (see 1.4.1)	I_{DRM2}		5	mA(pk)
5.	Gate-trigger voltage and current	4221	V_2 = V_B = 6 V dc; R_L = 50Ω; R_e = 20Ω maximum	V_{GT1} I_{GT1}	0.25	3 70	V dc mA dc
6.	Forward "on" voltage transfer ratio	4226	I_{TM} = 50 A(pk) (pulse); pulse width = 8.5 ms maximum; duty cycle = 2 percent maximum	V_{TM}		2.0	V(pk)

1/ The electrical measurements for table IVb (JAN and JANTX) of MIL-S-19500 are as follows:

Subgroups 2, 3, and 6, see table II herein, steps 1, 2, 3, 4, 5, and 6.

2/ The electrical measurements for table V of MIL-S-19500 are as follows:

Subgroups 2, 3, and 6, see table II herein, steps 1, 2, 3, 4, 5, and 6.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1).

b. Lead finish as specified (see 3.3.1).

c. Type designation and product assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue because of the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI, AV, SM
Navy - AS, CG, MC, SH
Air Force - 19, 85, 99
DLA - ES

Preparing activity:
Navy - EC

Agent:
DLA - ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS *p350Y

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/108E	2. DOCUMENT DATE (YYMMDD) 30 August 1993

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, THYRISTORS (CONTROLLED RECTIFIERS), SILICON, TYPES 2N682, 2N683, 2N685 THROUGH 2N692, 2N5206, JAN, AND JANTX

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 513-296-6048 (2) AUTOVON 986-6048
c. ADDRESS (Include Zip Code) DESC-ECT Dayton, OH 45444-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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